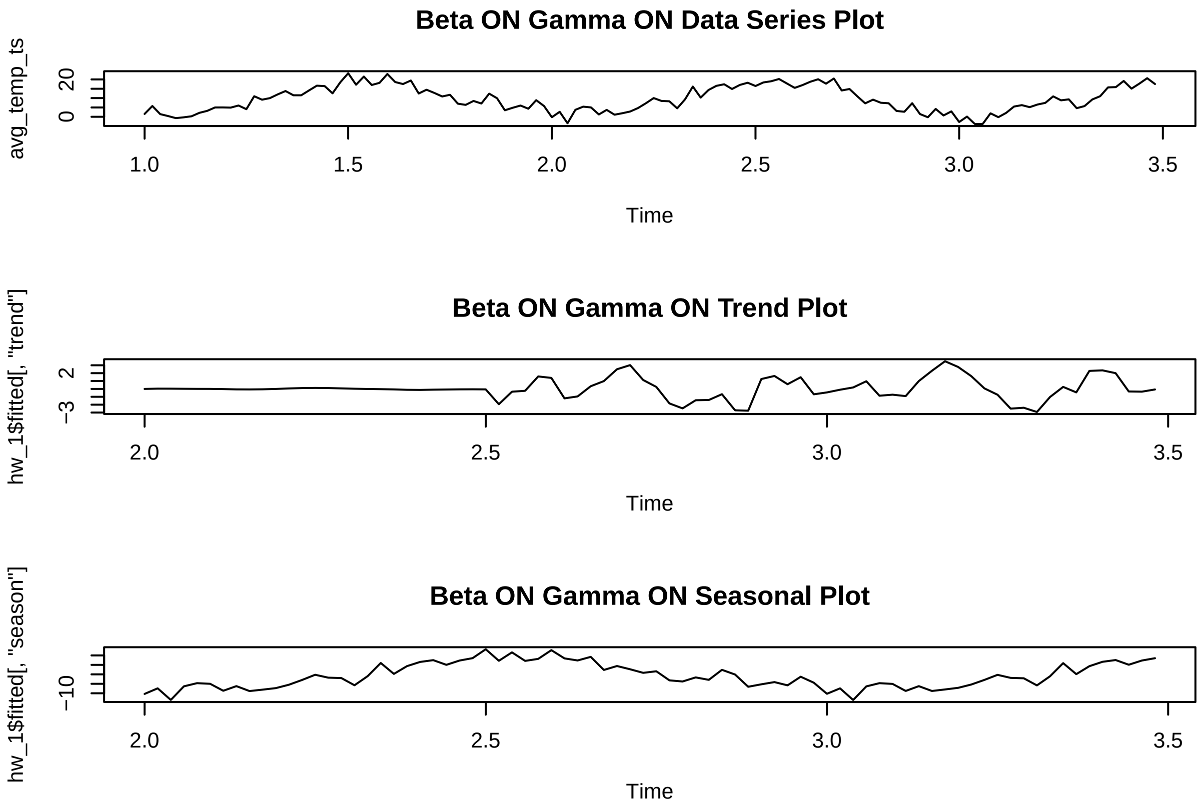
Time Series Forecasting Report

2026-03-07

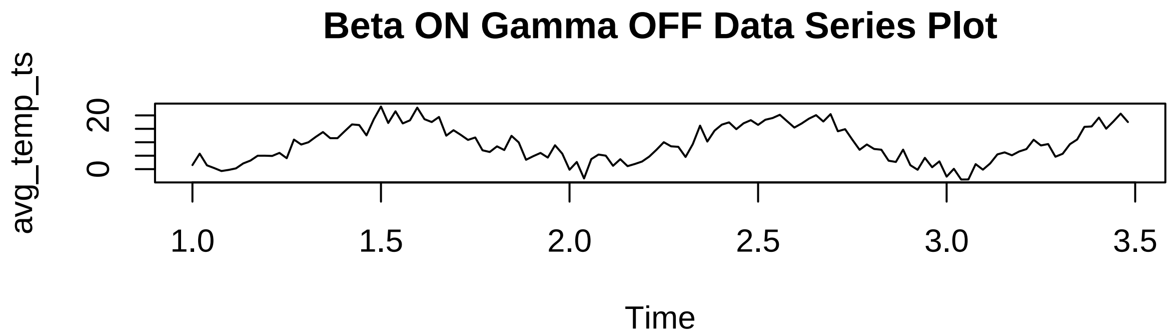
```r
# Beta OFF Gamma OFF
hw_4 <- HoltWinters(avg_temp_ts, beta = FALSE, gamma = FALSE)
```

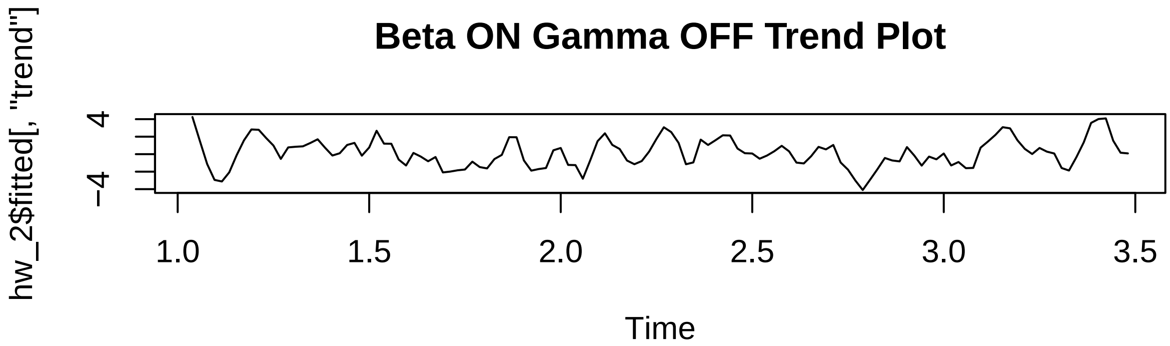
Plots

```r
# hw_1 plot
par(mfrow = c(3,1))
plot(avg_temp_ts, main ="Beta ON Gamma ON Data Series Plot")
plot(hw_1$fitted[,"trend"], type="l", main ="Beta ON Gamma ON Trend Plot")
plot(hw_1$fitted[,"season"], type = "l", main = "Beta ON Gamma ON Seasonal Plot" )
```

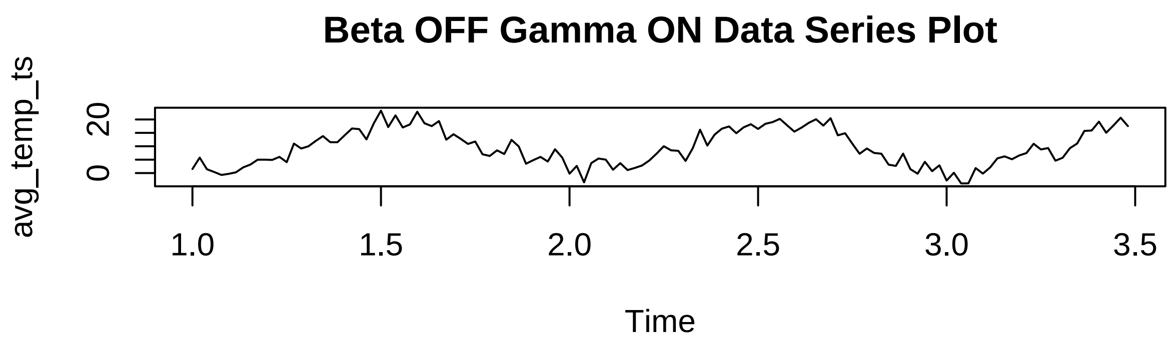


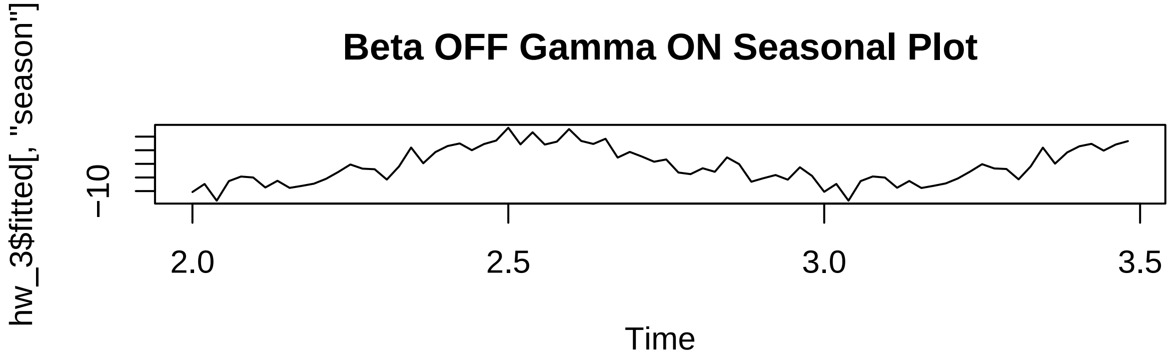

```r
# hw_2 plot
par(mfrow = c(2,1))
plot(avg_temp_ts, main = "Beta ON Gamma OFF Data Series Plot")
plot(hw_2$fitted[,"trend"], type = "l", main = "Beta ON Gamma OFF Trend Plot")
```



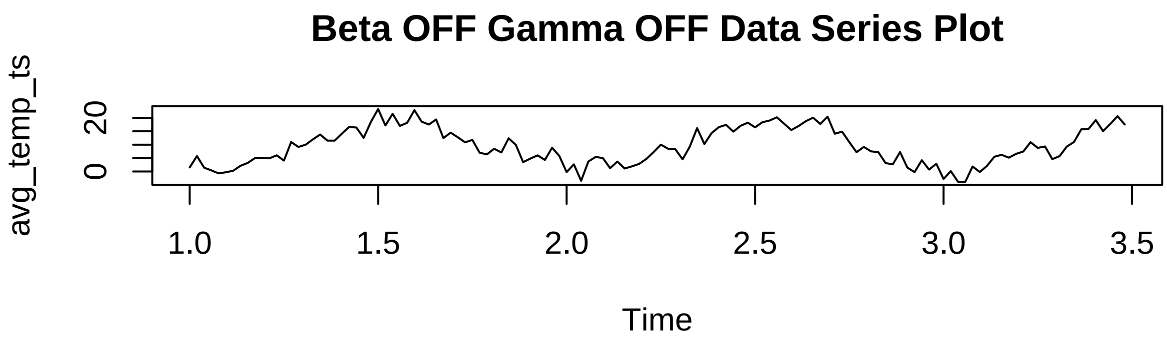



```r
# hw_3 plot
par(mfrow = c(2,1))
plot(avg_temp_ts, main = "Beta OFF Gamma ON Data Series Plot")
plot(hw_3$fitted[, "season"], type = "l", main = "Beta OFF Gamma ON Seasonal Plot")
```





```r
# hw_4 plot
plot(avg_temp_ts, main = "Beta OFF Gamma OFF Data Series Plot")
```

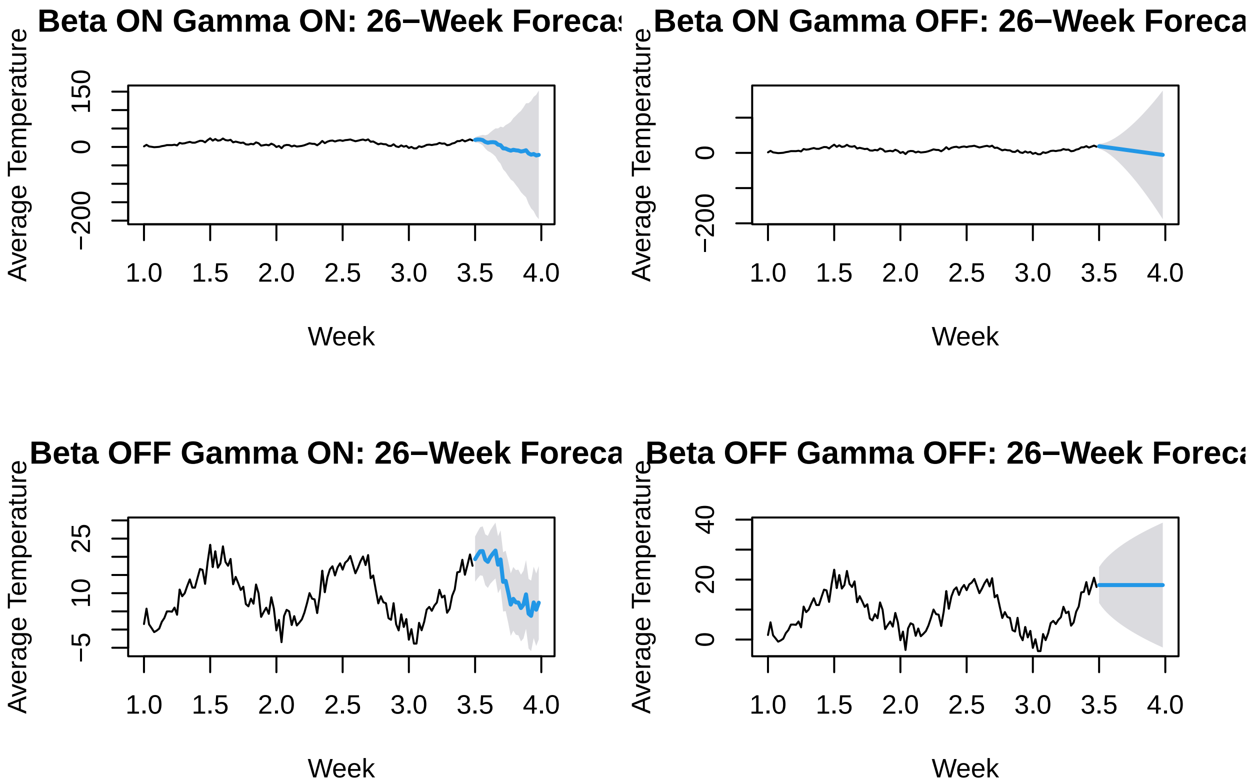


26-Week Sample Forecast

```r
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.5.2
```

```r
# 26-week out-of-sample forecasts with 95% confidence bands
fc_1 <- forecast(hw_1, h = 26, level = 95)
fc_2 <- forecast(hw_2, h = 26, level = 95)
fc_3 <- forecast(hw_3, h = 26, level = 95)
fc_4 <- forecast(hw_4, h = 26, level = 95)

# Plot all four forecasts
par(mfrow = c(2,2))

plot(fc_1,
     main = "Beta ON Gamma ON: 26-Week Forecast",
     xlab = "Week", ylab = "Average Temperature")

plot(fc_2,
     main = "Beta ON Gamma OFF: 26-Week Forecast",
     xlab = "Week", ylab = "Average Temperature")

plot(fc_3,
     main = "Beta OFF Gamma ON: 26-Week Forecast",
     xlab = "Week", ylab = "Average Temperature")

plot(fc_4,
     main = "Beta OFF Gamma OFF: 26-Week Forecast",
     xlab = "Week", ylab = "Average Temperature")
```




```
"We compared the forecasts from the four models for the next 26 weeks.
Some models produced very large and unstable forecasts, which suggests they may
not fit the temperature data well. The models with more stable forecasts are
likely better for predicting future temperatures."
```

```
## [1] "We compared the forecasts from the four models for the next 26 weeks. \nSome models produced ve
```